Fundamentals of Machine Learning

Tutorial 1, Extra Practice: Problems

From learning problems to probabilistic classification. Same skills as Tutorial 1, asked differently: reverse-engineering, error-spotting, fitting from raw data, proofs, tool choice, evasion.

How to use this set

This set covers the same skills as Tutorial 1 but asks for them in different ways; the problems are not the tutorial with new numbers. The tutorial hands you costs, parameters and a protocol and asks you to compute. Here you work backwards from a decision rule to the costs behind it, find the mistakes in someone else's solution, fit the classifiers yourself from raw data rows, pick an operating point from a validation table, prove the facts everything rests on, choose the right tool for a situation, and derive when a detector can be evaded. Part B is twelve true-or-false statements that each need a one-line justification.

Each problem opens with a blue box holding the setup, then a yellow box listing the tools you need with a pointer [one-read §x] to the one-read document. Work on paper first; the same set with full solutions is *ML_Tutorial1_Extra_Practice_Solutions.pdf*. Difficulty: ⋆ direct application, ⋆⋆ several steps, ⋆⋆⋆ derivation or proof. All logarithms are natural.

Contents

1 Part A: Problems **1**

1.1 Problem 1: Reverse-engineer the decision rule . . . 1
1.2 Problem 2: Find the mistakes . . . 2
1.3 Problem 3: From raw rows to three classifiers . . . 3
1.4 Problem 4: Choosing the operating point from a validation table . . . 3
1.5 Problem 5: Five facts the tutorial relies on . . . 4
1.6 Problem 6: Which tool, which protocol . . . 5
1.7 Problem 7: Evading the detector . . . 5

2 Part B: True or false, with one line of justification **6**

1 Part A: Problems

1.1 Problem 1: Reverse-engineer the decision rule

Problem 1 (⋆⋆) 15 minutes

An intrusion detection system was deployed by a team that has since left. Its documentation says only: "raise an alert when the model's attack posterior is at least 0.125". Analysts estimate that handling a false alert costs about one hour of work; take that hour as the unit of cost.

Tools you need

Conditional risk $R(a \mid x) = \sum_y c(a, y)\, p(y \mid x)$ and the Bayes rule [one-read §2.2]. The threshold formula $t = C_{FP}/(C_{FP} + C_{FN})$ assumes correct decisions cost 0.

(a) Assuming correct decisions cost nothing, what cost did the team implicitly assign to a missed attack?

(b) Correct decisions are not free: a true alert still costs analyst time. Take the general cost table below and derive the threshold on $p = p(\text{attack} \mid x)$. Then show that the table $a = 1$, $b = 2$, $M = 9$ gives the same threshold as (a), and explain why the two cost tables cannot be told apart from the rule alone.

	truth: benign	truth: attack
ignore	0	M
alert	a	b

(c) A third action, *block*, is added: it costs 5 on a benign flow (an outage) and 0.5 on an attack (the attack is stopped but still has to be investigated). With $a = 1$, $b = 2$, $M = 9$, derive the three decision regions.

(d) The analysts can handle at most 120 alerts per day. Of the 10,000 daily flows, 40 have posterior ≥ 0.5, another 80 have posterior in $[0.2, 0.5)$, and another 200 have posterior in $[0.125, 0.2)$. How many alerts does the cost-optimal rule of (a) produce? Propose a rule that respects the budget, and quantify what it gives up.

1.2 Problem 2: Find the mistakes

Problem 2 (⋆⋆) 15 minutes

A student was given the following exercise. *A logistic model was trained on data with attack prior* $\pi_t = 0.25$ *and is deployed where* $\pi_d = 0.05$*; assume prior shift and a calibrated training posterior. For one flow,* $r_t(x) = 0.8$*. Costs:* $C_{FP} = 2$, $C_{FN} = 6$*. On a test set of* 1,000 *flows with* 50 *attacks the model gives* $TP = 40$ *and* $FP = 60$*. Standardisation of the features was fitted on training and test data together. Decide what to do with the flow and assess the model.*
Their solution, numbered claim by claim:

(1) The cost threshold is $t = C_{FN}/(C_{FP} + C_{FN}) = 6/8 = 0.75$.
(2) Correcting for prior shift, $r_d(x) = r_t(x) \times \pi_d/\pi_t = 0.8 \times 0.2 = 0.16$.
(3) Since $0.16 < 0.75$, the flow is declared benign.
(4) Precision $= TP/(TP + FN) = 40/50 = 0.80$ and recall $= TP/(TP + FP) = 40/100 = 0.40$.
(5) $TN = 1000 - 50 - 60 = 890$, so accuracy $= (40 + 890)/1000 = 0.93$, which beats the all-benign baseline of 0.50; the model is useful.
(6) Fitting the standardisation on all the data is not leakage, because standardisation uses no labels.
(7) The threshold 0.75 is confirmed by the fact that it also gave the best F_1 on the test set.

Tools you need

Threshold [one-read §2.2]; prior-shift correction [one-read §4.7]; precision, recall, baselines, leakage, test-set reuse [one-read §3.5, 3.6].

(a) Go through the seven claims. For each, say whether it is right or wrong; if wrong, give the corrected statement or value.

(b) Write the correct final answer in two sentences.

1.3 Problem 3: From raw rows to three classifiers

Problem 3 ($\star\star\star$) **25 minutes**

Eight labelled hosts, two features each (x_1 = distinct internal hosts contacted per minute, x_2 = failed logins per minute, both already on comparable scales):

host	h_1	h_2	h_3	h_4	h_5	h_6	h_7	h_8
x_1	1	2	1	2	4	5	4	5
x_2	1	1	3	3	2	2	4	4
y	0	0	0	0	1	1	1	1

A new host has $x = (3, 4)$.

Tools you need

KNN and leave-one-out [one-read §2.4]; LDA with a shared covariance estimated by pooling, $\hat{\Sigma} = \frac{1}{2}(\hat{\Sigma}_0 + \hat{\Sigma}_1)$ with $\hat{\Sigma}_c = \frac{1}{n_c}\sum_{i \in c}(x_i - \hat{\mu}_c)(x_i - \hat{\mu}_c)^\top$ [one-read §4.4]; Gaussian Naive Bayes = one variance per feature per class [one-read §4.5].

(a) Classify x with 1-NN, 3-NN and 5-NN under Euclidean distance. Explain why 4-NN is ambiguous here.

(b) Compute the leave-one-out error of 1-NN on the eight training hosts. Does the result mean 1-NN generalises well here?

(c) Fit LDA: compute the class means, the pooled covariance, and the boundary with equal priors. Classify x and give its posterior.

(d) The boundary's normal vector is $(12, 1)$: almost vertical, although the means differ in both coordinates. Explain, and contrast with the Euclidean KNN of (a).

(e) Now fit Gaussian Naive Bayes (one variance per feature and per class). Show that it gives exactly the same boundary as LDA on this data, and state what would have to change for the two to differ.

(f) Attacks are actually 10% of hosts. Redo the LDA classification of x with $\pi_1 = 0.1$. Then say which of the three methods you would trust least with eight training points, and why.

1.4 Problem 4: Choosing the operating point from a validation table

Problem 4 ($\star\star$) **20 minutes**

A scoring model was run on a validation set of 550 flows, 50 of them attacks. The scores are grouped into bins:

score bin	$[0.9, 1]$	$[0.7, 0.9)$	$[0.5, 0.7)$	$[0.3, 0.5)$	$[0, 0.3)$
attacks	20	15	10	4	1
benign	2	8	20	70	400

Costs: $C_{FP} = 1$, $C_{FN} = 10$.

Tools you need

Precision, recall, F_1, false-positive rate, ROC [one-read §3.6]; the cost threshold [one-read §2.2]; calibration [one-read §4.7].

(a) For each threshold $t \in \{0.9, 0.7, 0.5, 0.3, 0\}$ (alert if score $\geq t$), compute TP, FP, FN, precision, recall, F_1 and the false-positive rate.

(b) Sketch the ROC points. Which threshold maximises F_1, and which minimises the expected cost $C_{FP} \cdot FP + C_{FN} \cdot FN$? Why do they differ?

(c) The Bayes threshold for these costs is $1/11 \approx 0.09$, yet the best threshold on the scores was 0.5. Compute, for each bin, the empirical fraction of attacks and compare it with the bin's scores. Resolve the apparent contradiction.

(d) Use the empirical fractions to build a calibration map (bin $\to$ fraction) and apply it to a new flow with score 0.55. With calibrated outputs, what threshold would you now use, and does the decision for this flow change?

(e) At deployment attacks are 1% of flows rather than $50/550 \approx 9\%$. Which quantities in the table of (a) stay valid and which do not, assuming only the prior changes?

1.5 Problem 5: Five facts the tutorial relies on

Problem 5 ($\star\star\star$) 25 minutes

Prove the following, in the notation of the lectures. X is the input, $Y \in \{0, 1\}$ the label, $p(x) = P(Y = 1 \mid X = x)$, and $\ell(y, a)$ a loss for action a.

Tools you need

Tower property $\mathbb{E}[Z] = \mathbb{E}[\mathbb{E}[Z \mid X]]$; conditional risk [one-read §2.2]; empirical and population risk [one-read §3.1]; the logistic gradient [one-read §4.2].

(a) The Bayes rule is optimal. Let $a^\star(x) \in \arg\min_a \sum_y \ell(y, a)\, P(Y = y \mid X = x)$. Show that for every decision rule $a(\cdot)$, $\mathbb{E}[\ell(Y, a(X))] \geq \mathbb{E}[\ell(Y, a^\star(X))]$.

(b) The Bayes error. For the 0–1 loss, show that the minimal achievable error is $R^\star = \mathbb{E}[\min\{p(X),\ 1 - p(X)\}]$, that $R^\star \leq \frac{1}{2}$, and that $R^\star = 0$ if and only if $p(X) \in \{0, 1\}$ with probability 1.

(c) Precision depends on the prevalence. Let $\pi = P(Y = 1)$, $\rho = P(\text{alert} \mid Y = 1)$ (recall) and $\phi = P(\text{alert} \mid Y = 0)$ (false-positive rate). Show that $\text{precision} = \dfrac{\rho\pi}{\rho\pi + \phi(1 - \pi)}$ and that it is increasing in π when $\rho, \phi > 0$.

(d) Training error is optimistic. Let $\widehat{R}_n(\theta) = \frac{1}{n}\sum_i \ell(y_i, f_\theta(x_i))$ on an i.i.d. sample, $R(\theta) = \mathbb{E}[\ell(Y, f_\theta(X))]$, $\hat{\theta} \in \arg\min_\theta \widehat{R}_n(\theta)$ and $\theta^\star \in \arg\min_\theta R(\theta)$. Show that $\mathbb{E}[\widehat{R}_n(\hat{\theta})] \leq R(\theta^\star) \leq \mathbb{E}[R(\hat{\theta})]$.

(e) Logistic regression is calibrated on average. For $r_i = \sigma(\theta^\top x_i + \theta_0)$ trained by minimising the summed cross-entropy, show that at the optimum $\sum_i r_i = \sum_i y_i$, i.e. the predicted number of attacks equals the observed number. Why does this fail if the intercept θ_0 is omitted?

1.6 Problem 6: Which tool, which protocol

Problem 6 (⋆⋆) 15 minutes

For each situation choose one of: k-nearest neighbours, logistic regression, LDA, (Bernoulli or Gaussian) Naive Bayes, or a window/sequence detector feeding a second classifier. Give one reason for the choice, one risk to watch, and the right validation split.

(i) 300 labelled phishing emails and 2,000 legitimate ones; features are 5,000 binary word indicators.

(ii) Two million flows, 40 numeric and strongly correlated features; the output must be a probability that will be thresholded by costs that change per customer.

(iii) Two well-chosen features, abundant data, and a scatter plot showing attacks as a small island inside the benign region.

(iv) Several features are frequently unavailable at decision time (a sensor is off); the model must still answer.

(v) The team must flag flows that look like *nothing* seen in training, not just flows that look like known attacks.

(vi) Each event looks innocuous; the attack is visible only as a pattern across many events from one source.

Tools you need

Local versus parametric [one-read §2.4, 2.5]; discriminative versus generative and their trade-offs [one-read §4.6]; handling missing and unfamiliar inputs [one-read §4.8]; splits and leakage [one-read §3.6].

(a) Give your six answers.

(b) Situation (ii) again, but the customer now also wants to know "which of the 40 features made this flow suspicious". Does the choice change?

1.7 Problem 7: Evading the detector

Problem 7 (⋆⋆⋆) 20 minutes

Two detectors watch failed login attempts from one source. Each attempt carries a score s from a per-attempt classifier.

- **Window detector:** alert when at least $k = 4$ attempts fall in the half-open window $(t - W,\ t]$ ending at the current attempt, with $W = 30$ s.
- **Decaying accumulator:** $A(t_i) = A(t_{i-1})\, e^{-(t_i - t_{i-1})/\tau} + s_i$, $A = 0$ initially, alert when $A \geq \theta$; here $\tau = 30$ s and $\theta = 0.55$.

A patient attacker sends one attempt every Δ seconds, each with score $s = 0.2$.

Tools you need

Window features and accumulation [one-read §4.8]; geometric series $\sum_{i=0}^{n-1} q^i = (1 - q^n)/(1 - q)$.

(a) Show that the window detector alerts if and only if $\Delta < W/(k-1)$, and compute the bound.

(b) For the accumulator with periodic attempts, write $q = e^{-\Delta/\tau}$. Show that $A_n = s\,\dfrac{1-q^n}{1-q}$ after n attempts, that A_n increases towards $A^\star = s/(1-q)$, and that the detector eventually alerts iff $A^\star > \theta$, i.e. iff $\Delta < \tau \ln \dfrac{1}{1-s/\theta}$ (assume $s < \theta$).

(c) Compute the bound in (b) for $s = 0.2$, $\theta = 0.55$, $\tau = 30$. Which attackers does the accumulator catch that the window detector misses?

(d) For $\Delta = 5$ s, how many attempts does the accumulator need before alerting, and at what time? Compare with the window detector.

(e) A legitimate but clumsy host produces a failed attempt every 5 s with score 0.05; a legitimate automation script with a stale password produces one every 5 s with score 0.10. Does the accumulator raise a false alert on either? What does this say about the choice of τ?

(f) The attacker learns the detector's parameters. What is the cheapest change to their behaviour, and what does that imply for how such thresholds should be managed?

2 Part B: True or false, with one line of justification

Decide each statement and justify it in one line. A bare "true" or "false" earns nothing.

(1) If two flows have identical features and different labels, no amount of additional training data can bring the error on such flows to zero.

(2) Lowering the alert threshold never decreases recall and never increases precision.

(3) In LDA with a shared covariance, changing the class priors rotates the decision boundary.

(4) With Laplace smoothing, Naive Bayes can still output a posterior of exactly 0 for some class.

(5) Of two logistic models fitted to the same data, the one with the lower training cross-entropy also has the lower training error rate.

(6) For the 0–1 loss, the Bayes error is at most 1/2.

(7) If the best of 100 configurations is chosen by validation error, that configuration's validation error is an unbiased estimate of its population error.

(8) For a logistic model, the rule "alert if $r(x) \geq t$" is the same as "alert if $s(x) \geq \ln \frac{t}{1-t}$".

(9) The prior-shift correction can change which of two flows has the higher posterior.

(10) Standardising the test features using the mean and standard deviation computed on the training set is leakage.

(11) A 1-NN classifier always has zero error on its own training set.

(12) A detector that alerts on four failed attempts in 30 seconds can be replicated by a per-attempt classifier with a suitably chosen threshold.